

Question:

A new arcade opens up and Jack decides to play with his 3 friends. Jack can play a game with 1 quarter for 20 minutes. Two of his friends are significantly worse than him and can only play half as long. One of them is significantly better and can play for 1.5 times as long. They play for 4 hours. How much money is used?

Reasoning + Answer:

<think> we know jack plays for 20 minutes with 1 quarter so we can write down his time like this
so we know jack plays for 20 minutes then we know two of his friends play half as long so we do
20 minutes split in half so we do 20 split in half which equals 10 so now we know two of his friends
play for 10 minutes each then we know one of his friends can play 1 and a half times longer than
jack so we take jack's time which was 20 and do 20 times 1 and a half which equals 30 so
now we know one of his friends plays for 30 minutes now we know jack plays for 20 minutes two
friends play for 10 minutes each and one plays for 30 minutes so now we add all these times up
so we do 20 minutes for jack then 10 minutes for one friend then 10 minutes for another friend
then 30 minutes for the best friend so we do 20 plus 10 plus 10 plus 30 which equals 70
so now we know in total they play for 70 minutes but we know they played for 4 hours and
there's 60 minutes in an hour so we do 4 hours times 60 which equals 240 so now we know
they played for 240 minutes in total so now we know they played for 240 minutes in total and
we know they played for 70 minutes so we do 240 take away 70 which equals 170 so now
we know they played for 170 extra minutes so now we know they played for 170 extra minutes
and we know jack can do 20 minutes per quarter so we do 170 split by 20 which equals 8 with
a remainder of 10 so now we know they used 8 quarters but we have 10 minutes leftover
which means they need an extra quarter for the remaining 10 minutes so now we add the 8
quarters and the 1 extra quarter which equals 9 so now we know 9 quarters were used </
think><answer>9</answer>

